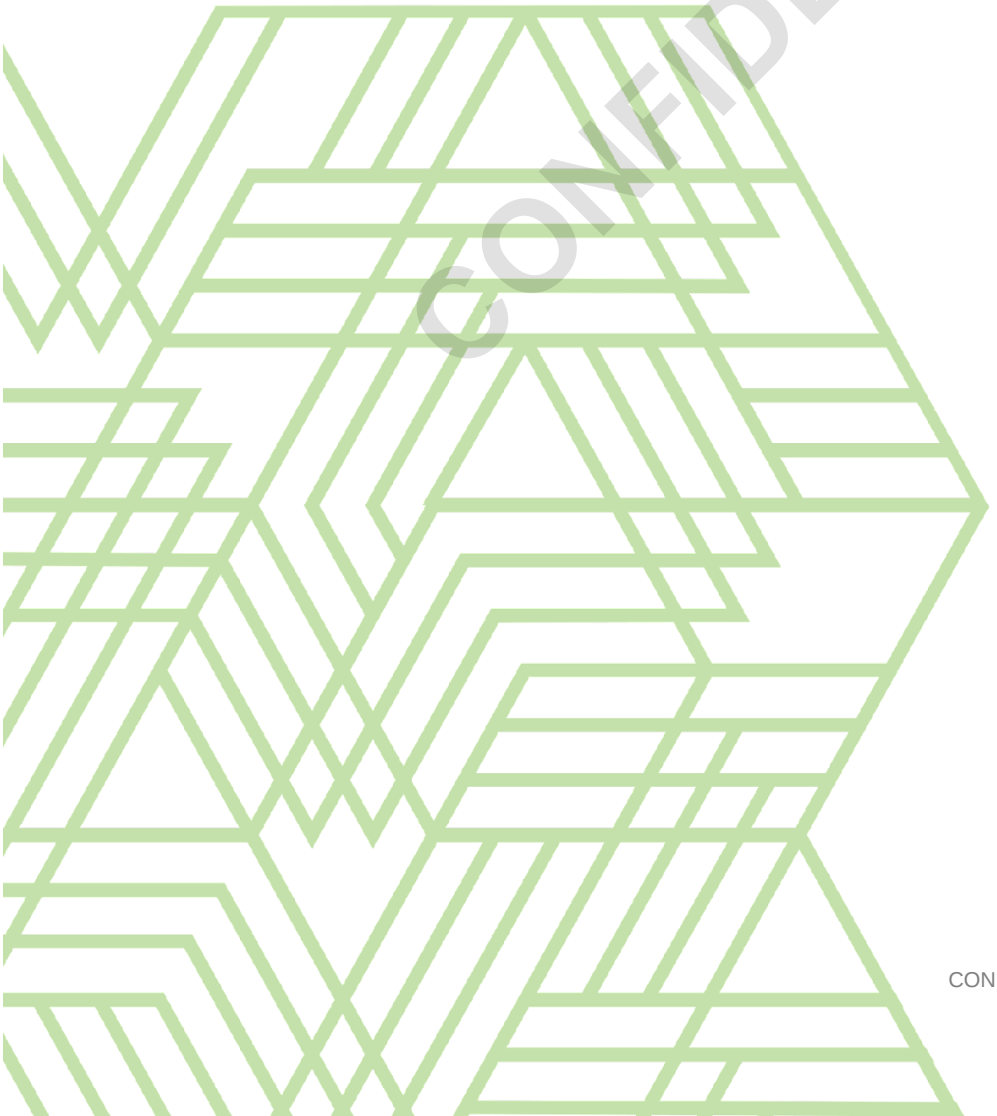
Atmosic SDK 5.3.0

Release Notes

SUMMARY: This document provides the release notes for Atmosic SDK 5.3.0 and associated tools and utilities enabling design and development of solutions based on ATM2, ATM3, or ATM33/e Wireless SoC Series. It lists the new features and known issues for this SDK release.

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1. Purpose

This release note provides details on new features/fixes/known issues as part of the Atmosic Software Development Kit (SDK). This Atmosic SDK 5.3.0 supports Evaluation Kit (EVK) platform based on following devices:

ATM2 Series	ATM3 Series	ATM33/e Series
ATM2201	ATM3201	ATM3330e 7x7 mm QFN Package
ATM2202	ATM3202	ATM3330 7x7 mm QFN Package
ATM2221	ATM3221	ATM3325 5x5 mm QFN Package
ATM2231	ATM3231	
ATM2251		

1.1 Development Platforms, IDE and Tool Chains

This section listed the OS and debug tools supported by the ATM33/e series:

Category	Features Supported
Development Platforms	MacOS (x86, M1)
	Linux, Windows 10, Windows 11
IDE	Keil with ARM C++ compiler
	IAR Workbench
	SEGGER Embedded Studio
Debug Tools	J-Link Debugger (on-board for ATM33/e EVKs)
	OpenOCD debug
FW Update	OTA FW Update

Please visit Atmosic Support website for the following documents:

- ATM33/e Series SDK User Reference Manual
- ATM33/e Series EVK User Guide
- ATM33/e (Rev. B0) Series Release Notes
- ATM33/e Series Register Specification

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- ATM33/e Digital Biquad Filter Software Implementation Application Note
- Complete Doxygen-style reference of all APIs provided by this SDK
- Bluetooth LE functionality, including GAP, GATT, and various profiles

Additional references for developing applications using third party IDE Software:

- IAR Workbench User Guide
- Keil MDK Configuration Guide
- SEGGER Embedded Studio User Guide
- IDE Auxiliary Flash Programming Tool User Guide

Please submit a support request for access credentials to the Atmosic Support website.

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2. New Features / Functionality

2.1 SDK and Application Version

SDK and application versions are displayed as part of the boot up messages:

@009404ab SDK Version: 5.3.0

@00940504 APP Version: 0.0.0.9

- SDK version is located in version.h folder in <ROOT>/AtmosicSDK
- App version can be specified using APP_VERSION defined during make

2.2 Supported Devices

This release combines the support for all the Atmosic SoC Series - ATM2, ATM3, and ATM33/e.

ATM2		ATM3		ATM33/e
ATM2201-x0x	ATM2201-x1x	ATM3201-x0x	ATM3201-x1x	ATM3330e-5DCAQN
ATM2202-x0x	ATM2202-x1x	ATM3202-x0x	ATM3202-x1x	ATM3330-5DCAQN
ATM2221-x0x	ATM2221-x1x	ATM3221-x0x	ATM3221-x1x	ATM3325-5DCAQK
	ATM2231-x1x		ATM3231-x1x	
	ATM2251-x1x			

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2.3 Features

2.3.1 ATM2/ATM3

This section describes the new features for the ATM2/ATM3 series supported by the SDK 5.3.0 release.

Category	Features Supported
QSPI Flash	Support for Giantec QSPI Flash

2.3.2 ATM33/e

This section describes the list of features for the ATM33/e series supported by the SDK 5.3.0 release.

Category	Features Supported
AoA - Angle of Arrival (BT5.1)	AoA Transmitter - For RTLS Asset Tags
Bluetooth LE Core Features	Bluetooth LE Power Control
	Peripheral and Broadcaster role
	Central role (multiple master)
Bluetooth LE 5.2 Features	Enhanced Attribute Protocol (EATT)
Quuppa RTLS	Quuppa Tag support
PHY Rates	1 Mbps
	2 Mbps
	125 kbps / 500 kbps Coded PHY
Clocks	16 MHz
	32 MHz
	64 MHz
Internal RC	32 kHz Internal RC oscillator (RCOSC)
Audio Interfaces	PDM
Peripherals	I2C Master
	UART
	SPI Master
	PWM
	IR Tx Support (PWM FIFO based)
	GADC
	QSPI external flash support - data storage
Audio Codecs and PTT/HTT	ADPCM

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Category	Features Supported
Software support	Google 1.0 Voice Support
	mSBC (APIs)
	Programmable Filters/Equalizer, Bi-quad support
Power Management Unit	3 V Battery Extension mode (3 V Battery and Supercap)
	Storage Configuration/Type
	VBAT_LI LDO Enablement: Full operating range; Auto control
Harvesting Recharge (applicable to ATM3330e only)	Direct recharging from harvesting when VBAT_LI < 3.4 V
	HSC operation/recharge using discrete booster
	LiFePO4 battery 2.5 to 3.4 V (operation+recharging)
	Li-Ion Battery 4.2 V (operation+recharging)
Energy Harvesting (applicable to ATM3330e only)	Standalone Multi-cell PV support
	Standalone Single-cell PV support - 500 mV
	RF - 2.4 GHz
	RF - 914 MHz
	Non-concurrent harvesting (PV or RF)
	Concurrent harvesting without MPPT (multi-cell PV and RF)
	PV/RF charging indicator (charging/not-charging)
	Overvoltage protection of Battery/HSC
	Undervoltage/brownout protection of Battery/HSC
Security	AES-128/256 HW cryptographic engine
	SHA2-HMAC-256 HW cryptographic engine
	TRNG
	ECDSA for signing (based on MCU boot)
	Secure Debug (driver to lock down debug ports)
	ARM TrustZone

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Category	Features Supported
	Root-of-Trust
	Secure Watchdog
	Secure Boot (ECDSA signing/verification)
	Secure firmware upgrade with Anti-rollback

3. Major Changes / Bug Fixes

- Major changes and bug fixes in SDK 5.3.0 include:

  Added support for Windows 11 and MacOS M1 development platforms

  Each example folder no longer provides a built-in 3rd party IDE project. Users can generate it according to compilation options through the IDE Auxiliary Flash Programming Tool.

  - ATM2/ATM3
    - Support Bluetooth LE privacy in HID_remote example
  - ATM33/ATM33e
    - Functional support for operations from Li-Ion battery
    - Functional support for recharging the Li-Ion battery with harvested energy
    - Support for AES 128/256 HW engine and examples
    - SHA2 driver extended to support HMAC-256 Mode with examples
    - Secure Counters
    - MCUboot Anti-rollback support using Secure Counters
    - Secure watchdog support in MCUboot and SPE (bootloader)
    - Updated uECC library to improve the performance of Secure Boot ECDSA

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4. SDK Platform Working Directory

Following is a list of all Atmosic SDK platform working directories:

- For ATM2/ATM3 devices use the following directories based on the device revision being used:
  - platform/atm2/ATM22xx-x0x
  - platform/atm2/ATM22xx-x1x
  - platform/atm3/ATM32xx-x0x
  - platform/atm3/ATM32xx-x1x
- For ATM3330/ATM3330e devices use the directory:
  - platform/atm33/ATM33xx-5

4.1 Reference Examples and Applications

The reference examples and applications newly added with this release are:

- ATM2/ATM3:
  - Same as SDK 5.2.0 release
- ATM33/e:
  - BLE_att_client (enhanced to support EATT)
  - BLE_att_server (enhanced to support EATT)
  - BLE_atcmd
  - CGM_sensor
  - crypto_demo
  - GADC_test

5. ATM33/e Tx Power Settings

Tx output power settings default to 0 dBm or +4 dBm for the supported examples and applications listed in the Reference Examples and Applications section. To enable Tx output power higher than +4 dBm, please submit a support request for additional details.

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6. Limitations and Known Issues

Following limitations and known issues apply to this release for ATM2/ATM3 and ATM33/e devices.

6.1 ATM2/ATM3

Issue	HID_remote occasionally not able to reconnect to Pixel 4A [G-1982]
Impact	This is caused by a known issue of Android 13 that is assigned to AOSP developers: (https://issuetracker.google.com/issues/242755161). Pixel devices running Android 13 tend to be unstable and fail to reconnect bonded BLE peripherals. There is no effective WAR currently available for this issue.
Resolution	This is an Android 13 issue, and not related to the ATM2/ATM3.

Issue	HID_remote enabled Bluetooth LE privacy cannot reconnect to iPhone after OTA [G-2000]
Impact	Minor. Cannot reconnect to the iPhone.
Resolution	Users can turn the phone's BT radio OFF then turn it ON to make it work. To be fixed in future release

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6.2 ATM33/ATM33e

Issue	Cannot set BD address through AT command [P-1923]
Impact	Minor. This applies only for cases where the BD address is not programmed in the manufacturing floor. Typically, in most manufacturing setups SWD access is available to program this field.
Resolution	To be fixed in the next release.

Issue	ASSERT seen on system reset after flashing [P-1961]

Impact	Minor. This issue is seen intermittently (once in about 1.5 to 2 hours of continuous reboot testing. This issue may be seen when resetting the board after downloading new binary to the board
Resolution	To be fixed in the next release.

7. Resolved Issues

Following limitations and known issues are resolved in SDK 5.3.0 release.

7.1 ATM2/ATM3

Issue	ATM2/3 HCI UART Rx Overflow under stress
Impact	When stressing the HCI UART (HCI example) with the ATM2 FTDI UART board and PySerial, an RX overrun is detected in the uart0 driver. This issue was observed intermittently with the ota_demo application which pushes a significant amount of data over the FTDI UART board through the HCI interface.
Resolution	Resolved

7.2 ATM33/ATM33e

Issue	Undervoltage protection (brownout) support not enabled in current release
Impact	Cannot enable brownout protection
Resolution	Resolved

Issue	Assert with HCI app asserts after 15 minutes if no packets are sent to it [P-220]
Impact	Low. This issue can be avoided by sending packets to the application sooner than the 15 minute timeout period
Resolution	Resolved

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Issue	Failed to gather core dump after test failure [S-1472]
Impact	Moderate. Coredump collection is still being implemented
Resolution	Resolved

Issue	HID_remote application voice search with ADT3 may not work if enabling OTA and compiling ATVV version to use 0.4e [P-1502]
Impact	Moderate. Application will use ATVV version 1.0 by default.
Resolution	Resolved

Issue	HID_remote application voice search doesn't work with ADT3 after reconnection if enabling OTA [P-1574]
Impact	Moderate. There is no problem if not enabling OTA or after first pairing. ADT3 box will not do MTU exchange in the reconnection stage. If the box or TV will do MTU exchange, the voice search function can work.
Resolution	Resolved

Issue	I2C_client/server example doesn't work [P-1454]
Impact	Moderate. I2C master functionality is supported and available through other sensor based examples
Resolution	Resolved

Issue	Assert seen in HID Remote app during OTA process when paired with another ATM33 EVK board running HCI app configured for OTA [P-1527]
Impact	Low. This issue is only seen when running the HID remote with the BD_address set to 00-00-00-00-00-00
Resolution	Resolved

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Issue	Pinmux tool support for ATM3325 5x5 mm QFN package is not included [P-1232]
Impact	Moderate.
Resolution	Resolved

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Revision History

Date	Version	Description
November 9, 2022	1.00	Initial version created for SDK 5.3.0 release.

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